PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947—enacted by the Govt. of India)

FACULTY OF ARTS

SYLLABI

FOR

M.A. Economics Part-II
Examinations, 2012

--:o:--

© The Registrar, Panjab University, Chandigarh.
All Rights Reserved.

APPLICABILITY OF REGULATIONS FOR THE TIME BEING IN FORCE

Notwithstanding the integrated nature of a course spread over more than one academic year, the regulations in force at the time a student joins a course shall hold good only for the examinations held during or at the end of the academic year. Nothing in these regulations shall be deemed to debar the University from amending the regulations subsequently and the amended regulations, if any, shall apply to all the students whether old or new.

GUIDELINES FOR CONTINUOUS INTERNAL ASSESSMENT (20%) FOR REGULAR STUDENTS OF POST-GRADUATE COURSES (Annual System) *Except* M.Sc. (Mathematics)

(Effective from the First Year Admissions for the Academic Session 2004-2005)

1. The Syndicate has approved the following guidelines, mode of testing and evaluation including Continuous Internal Assessment of students :
   (i) Terminal Evaluation : 80 %
   (ii) Continuous Assessment : 20 %
   (iii) Continuous Assessment may include written assignment, snap tests, participation in discussions in the class, term papers, attendance etc.
   (iv) In order to incorporate an element of Continuous Internal Assessment of students, the Colleges/Departments will conduct three tests as quantified below :

(a) Three Tests :				
I	25	} 2 best tests	=	50
II	25			
III	25			
(b) Snap Tests			:	15
(c) Participation in class discussions			:	10
(d) Term Paper/s			:	15
(e) Attendance			:	10
		Total	**:**	**100 reduced to 20**

2. Weightage of 2 marks for attendance component out of 20 marks for Continuous Assessment shall be available only to those students who attend 75% and more of classroom lectures/seminars/ workshops. The break–up of marks for **attendance component** for theory papers shall be as under :

Attendance Component		*Mark/s for Theory Papers*
(a)	75 % and above upto 85 %	: 1
(b)	Above 85 %	: 2

3. It shall **not be compulsory** to pass in Continuous Internal Assessment. Thus, whatever marks are secured by a student out of 20% marks, will be carried forward and added to his/her score out of 80 %, i.e. the remaining marks allocated to the particular subject and, thus, he/she shall have to secure pass marks both in the University examinations as well as total of Internal Continuous Assessment and University examinations.
4. Continuous Internal Assessment awards from the affiliated Colleges/Departments must be sent to the Controller of Examinations, by name, **two weeks before** the commencement of the particular examination on the *proforma* obtainable from the Examination Branch.

SPECIAL NOTE :

(i) The theory question paper will be of 80 marks and 20 marks will be for internal assessment.
(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.
(iii) In the case of Postgraduate Courses in the Faculties of Arts, Science, Languages, Education, Design & Fine Arts, and Business Management & Commerce (falling under the purview of Academic Council), where such a provision of Internal Assessment/Continuous Assessment already exists, the same will continue as before.
(iv) The marks obtained by a candidate in Continuous Internal Assessment in Postgraduate Classes from the admissions of 2004 will be shown separately in the Detailed-Marks-Card (D.M.C.).

PANJAB UNIVERSITY, CHANDIGARH

OUTLINES OF TESTS, SYLLABI AND COURSES OF READING IN ECONOMICS FOR M.A. PART II EXAMINATION, 2012

OUTLINES OF TEST

Compulsory Papers

Paper- I	**:**	Macro-Economic Theory	80 marks
Paper -II	**:**	Economics of Growth, Development and Planning	80 marks

Optional Papers

Papers III & IV - Any **two** from the following :

Option (i)	Econometrics	80 marks
Option (ii)	Mathematical Economics	80 marks
Option (iii)	International Economics	80 marks
Option (iv)	Economics of Public Finance	80 marks
Option (v)	Economics of Money & Banking	80 marks
	OR	
	Economics of Financial Markets and Institutions	80 marks
Option (vi)	Political Economy	80 marks
Option (vii)	Computer Applications in Economic and Statistical Analysis	80 marks
Option (viii)	Economics of Agriculture or Economics of Industry – whichever the student has not offered in M. A. – I. The syllabi in this option will be the same as prescribed in M.A.-I.	80 marks
Option (ix)	Economics of Environment	80 marks
Option (x)	Dissertation	100 marks

Note :

1. Only those candidates who have obtained at least 55% (fifty five percent) marks in the M. A. Part I examination can opt for Option (x) Dissertation.

2. Only Regular students will be allowed to offer Option (vii) and the prescribed course of attendance in practical in this paper will also be compulsory.

SYLLABI AND COURSES OF READING FOR M.A. PART II ECONOMICS

PAPER I : MACRO-ECONOMIC THEORY

100 Hours

Macroeconomics or aggregative economies analysis establishes the functional relationship between the large aggregates. The aggregate analysis has assumed such a great significance in recent times that a prior understanding of macroeconomic theoretical structure is considered essential for the proper comprehension of the different issues and policies. Macroeconomics now is not only a scientific method of analysis; but also a body of empirical economic knowledge.

The paper entitled "Macro-Economic Theory" equips the students at the postgraduate level to understand systemic facts and latest theoretical developments for empirical analysis.

The time-tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars.

The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Instructions for the Paper-setters and candidates :

(i) The theory question paper will be of 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

The paper setter must put note (ii) in the question paper.

The syllabus has been divided into four units.

There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 10 short answer type questions carrying 20 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice of attempting one question from each Unit – 4 in all. Each question will carry 15 marks.

UNIT-I

Consumption Function : Absolute Income Hypothesis, Relative Income Hypothesis, Permanent Income and Life-Cycle Hypotheses. Consumption under Uncertainty – Modern Approach.

Investment Function : Keynesian approach. Accelerator Theory, Neo-classical theory of investment and Tobin's Q theory of investment.

UNIT–II

Demand for Money : Quantity theory of money, Keynesian Approach, Baumol and Tobin's Contributions, Friedman's restatement of Quantity Theory of money.

Supply of Money : Definition of money and its importance in macro-economics. Money multiplier and credit creation by commercial banks.

Derivation, properties and shifts of IS and LM curves and simultaneous equilibrium in money and product markets. Effects of monetary and fiscal policies under different cases in the IS-LM framework.

UNIT–III

Demand and Supply of Labour : Classical and Keynesian views.

Keynesian and classical models of income and employment determination. Wage-price flexibility and automatic full employment – Classical versus Keynesian approach.

Inflation : Its effects, Kaldorian theory of distribution. Quantity theory view, Keynes' view. Phillips curve analysis, Long run Phillips curve, monetarist theory and structuralist theory of inflation.

Trade Cycle Theories : Accelerator – Multiplier Interaction models, Kaldor model.

UNIT–IV

Balance of Payments : Current and Capital Accounts, Mundell–Flemming open economy model, Disequilibrium in balance of payments and its consequences, Balance of payments, adjustment policies in fixed and flexible exchange rates.

Monetary and Fiscal Policies : Objectives, conflicts among objectives, Mundell Model, Swan Model.

Recent Developments : Monetarism, New-Classical Macro Economics : Rational Expections revolution and Real Business Cycle theory. New Keynesian Macro economics.

BOOKS RECOMMENDED :

Ackley, G. (1978) : Macroeconomics : Theory and Policy, Macmillan, New York.

Aschheim, J. and Hsiech, C. (1969) : Macroeconomics : Income and Monetary Theory, Columbus, Ohio, Charles E. Merrill Publishing House.

Blanchard, O. J. and Fisher, S. (1989) : Lectures in Macroeconomics, MIT Press, Cambridge.

Branson, W. H. (1994) (3rd ed.) : Macroeconomic Theory and Policy, Harper and Row, New York.

Branson, W. H. & Litvack, J. M. (1976) : Macroeconomics, Harper and Row, New York.

Brooman, F. S. (1975) (5th edn.) : Macro Economics, George Allen and Unwin Ltd., London.

Burrow, P. & Hitiris, T. (1974) : Macro Economic Theory : A Mathematical Introduction, John Wiley & Sons, New York.

Curwen, P. J. (1976) : Inflation, Macmillan and Co., London.

Dernberg, T. F. and Mc Dugall, D. M. (1976) : Macro Economic Analysis (4th Ed.), McGraw Hills, New York.

Dornbusch, R. (1980) : Open Economy, Macro Economics, Basic Books, New York.

Dornbusch, R. & F. Stanley (1997) : Macroeconomics, McGraw Hill, Inc., New York.

Duesenberry, J. S. (1949) : Income, Saving and the Theory of Consumer Behaviour, Harvard University Press, Harvard.

Eugene, Diulio (1997) : Macroeconomics, Schaum's Outlines of Theory and Problems, McGraw Hill.

Friedman, M. (1970) : A Theory of the Consumption Function, Oxford and IBH.

Friedman, M. (1956) : Studies in the Quantity Theory of Money, The University of Chicago Press, Chicago.

Frisch, H. (1983) : Theories of Inflation, Cambridge University Press, Cambridge.

Gordon, R. and S. G. Harris (1998) : Macroeconomics, Addison Wesley, New York.

Hagger, A. J. (1977) : Inflation : Theory and Policy, Macmillan, London.

Hall, R. E. & J. B. Taylor (1986) : Macroeconomics, W.W. Norton, New York.

Heijdra, B. J. & V. P. Frederick (2001) : Foundations of Modern Macroeconomics, Oxford University Press, New Delhi.

Hicks, J. R. (1950) : A Contribution to the Theory of Trade Cycles, Clarendon Press, Oxford.

Hicks, J. R. (1974) : The Crisis in Keynesian Economics, Oxford University Press, New Delhi.

Jha, R. (1999) : Contemporary Macroeconomic Theory and Policy, New Age International (P.) Ltd., New Delhi.

Keynes, J. M. (1936) : The General Theory of Employment, Interest and Money, Macmillan, London.

Laidler, D. E. W. (1977) : Demand for Money : Theory and Evidence, Dum-Don Valley, New York.

Levacic, R. and R. Rebman (1986) : Macroeconomics : An Introduction to Keynesian and Neo-classical Controversies, Macmillan, London.

Lucas, R. (1981) : Studies in Business Cycle Theory, MIT Press, Cambridge, USA.

Mankiw, N. G. and D. Romer (eds.) (1991) : New Keynesian Economics (2 Vols.), MIT Press, Cambridge.

Mathews, R. C. O. (1964) : The Trade Cycle, Cambridge Univ. Press.

Mckinen, G. E. (1978) : Money, The Price Level and Interest Rates, Prentice Hall of India, New Delhi.

Mueller, M. G. (1966) (Ed.) (3rd ed.) : Readings in Macroeconomics, Holt Rinehart and Winston, New York.

Patinkin, D. (1965) : Money, Interest and Prices (2nd ed.), Harper and Row, New York.

Peston, M. H. (1982) : Theory of Macroeconomic Policy, Heritage Publishers.

Rakshit, M. (1998) : Studies in the Macroeconomics of Developing Countries, Oxford University Press, New Delhi.

Romer, D. L. (1996) : Advanced Macroeconomics, McGraw Hill Co. Ltd., New York.

Scarfe, B. L. (1977) : Cycles, Growth and Inflation, McGraw Hill Co. Ltd., New York.

Shapiro, E. (1996) : Macro-economic Analysis (5th ed.), Galgotia Publications, New Delhi.

Surrey, M. J. C. (ed.) (1976) : Macroeconomics Themes, Oxford University Press, Oxford.

Smith, W. L. and Teigen, R. L. (Ed.) (1970) : Readings in Money, National Income and Stabilization Policy, Richard D. Irwin.

Taylor, L. (1983) : Structuralist Macroeconomics, Basic Books, New Longman.

Turnovsky, S. J. (1977) : Macroeconomic Analysis and Stabilization Policy, Cambridge University Press, Cambridge, USA.

American Economic Association and The Royal Economic Society (1965) : A Survey of Economic Theory, Vols.1, 2 & 3.

PAPER II : ECONOMICS OF GROWTH, DEVELOPMENT AND PLANNING

100 Hours

As a sequel to the post-second war developments, the study of Economic Development, gained impetus because three-fourths of humanity was experiencing wretched conditions of existence. There was a pressing need in those countries for uplifting their economic conditions by restructuring their economies to acquire greater diversity, efficiency and equity, in consonance with their priorities.

Since a variety of perspectives were available, the policy makers were eager to acquaint themselves with various policy options in their bid to re-construct their dilapidated economies. In addition, various international bodies were also keen to help and guide the laggards. Consequently, the study of development economics assumed greater significance.

In recent times, the resurgence of marketism and greater focus on areas like education, health, sanitation, energy and environment, and infrastructure development, hitherto relegated to the background, have reopened some of the old debates besides opening up new areas of investigation. Growth and Development economists are making earnest efforts at theorizing to break fresh grounds. Consequently, study of this discipline continues to be of prime importance.

Modules incorporated in this paper are devoted to the theories of growth and development, importance of agriculture, and the rationale and pattern of industrialization in developing countries. The other important issues in the context of development such as infrastructure-linkages, role of international trade, importance of economic policies and relevance of planning have been included in the modules of this paper.

The time-tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience.

The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Instructions for the Paper-setters and candidates :

(i) The theory question paper will be of 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.
The paper setter must put note (ii) in the question paper.

The syllabus has been divided into four units.

There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 10 short answer type questions carrying 20 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice of attempting one question from each Unit –4 in all. Each question will carry 15 marks.

UNIT-I

Modern Economic Growth – Rate, Structure and Spread : Economic growth of Nations in the Post-Industrial revolution era and Kaldor's stylised facts of growth.

*Economic Growth and Structural Change (*Hypothesis and empirical regularities relating to changes in the composition of output, occupational structure, demographic changes and income distribution—the Kuznet's curve).

*Theorising Growth (*The Models of Solow-Swan, Kaldor-Passineti, Robinson; Convergence/Divergence debate without reference to the recent thrust in the theory of international trade—the ideas of Arrow, Lucas and Romer—their Specific Models not to be derived).

UNIT–II

From Economic Growth to Economic Development :
Defining Development (Development, Growth and Change; Goulet's Sufficiency for all, and Sen's Development as Freedom).

Measuring Development (Income/Output based measures and their inadequacies; rationale, construct, and use of PQLI and HDI as indicators of development).

Population Growth and Economic Development (Their interaction; Spatial distribution of population and developing economies).

Economic Growth and the Issue of Sustainable Development (Resource depletion, resource augmentation, environmental consequences and sustainability).

UNIT–III

Theories of Development & Underdevelopment :
Explaining Underdevelopment (Myrdal's cumulative causation thesis, theories of Raul Prebisch and E. Emmanual).

Theories and Processes of Development (Classicals—only a synthesized version; Karl Marx and Joseph A. Schumpeter).

UNIT-IV

Strategies for Development :

The Dual-Economy (Models of W.A. Lewis, and Fei and Ranis; Harris & Todaro model of Rural-Urban migration).

Balanced and Unbalanced Growth :

Economic Isolation and Integration with the Global Market (Reliance on foreign-trade, international aid, FDI and MNCs).

BOOKS/JOURNALS RECOMMENDED :

A.P. Thirlwal (1999) : Growth and Development with Special Reference to Developing Economies, Macmillan, London.

A. K. Sen, (ed.) (1970) : Growth Economics, Penguin Books, London.

Amartya Sen (1999) : Development as Freedom, Oxford University Press, London.

Debraj Ray (1998) : Development Economics, Oxford University Press, Delhi.

Gerald M. Meier : Leading Issues in Economic Development, Oxford University
James E. Rauch (2000) Press, New York.

Gustav Ranis & John C.H. Fei : "A Theory of Economic Development", American Economy
(1961) Review, September, pp. 533-65.

Hollis Chenery & Moises Syrquin : Patterns of Development : 1950-70, Oxford University Press.
(1975)

Kenneth Arrow (1962) : The Economic Implications of Learning by Doing Review of Economic Studies, June, pp. 155-173.

Michael P. Todaro (1997) : Economic Development, Addison-Wesley, Mass, ISE.

Pan A. Yotopoulos : Economics of Development–Empirical Investigations, Harper
Jeffrey B. Nugent (1976) & Row, New York.

Paul, M. Romer (1986) : Increasing Returns and Long Run Growth, Journal of Political Economy, October.

----------(1990) : Endogenous Technological Change, Journal of Political Economy, October.

Robert Lucas (1988) : On the Mechanics of Economic Development, Journal of Monetary Economics, July.

S. Kuznets (1966) : Modern Economic Growth, Rate, Structure and Spread, Vakils, Feffer and Simons Private Limited, Bombay.

Sukhmoy Chakravarty (1982) : Alternative Approaches to Theories of Economic Growth : Marx, Marshall and Schumpeter, R.C. Dutt Lectures on Political Economy, 1980, Centre for Studies in Social Sciences, Calcutta, Orient Longman.

Theodore W. Schultz (1981) : Investing in People, Hindustan Publishing Corporation (India), Delhi.

Thorvaldur Gylfason (1999) : Principles of Economic Growth, Oxford University Press.

PAPERS III & IV OPTION (i) : ECONOMETRICS

100 Hours

Application of economic theory need a reasonable understanding of economic relationships and relevant statistical methods. The econometric theory thus becomes a very powerful tool for understanding of applied economic relationships and for meaningful research in economics. This paper accordingly is devoted to equip the students with basic theory of econometrics and relevant applications of the methods. The topics covered in the course include various problems faced in estimation of both single equations and simultaneous equations models.

Besides the time-tested method of imparting verbal instructions through lectures, description and derivation of econometric models would be demonstrated by extensive use of blackboard. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars.

The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Instructions for the Paper-setters and candidates :

(i) The theory question paper will be of 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.
The paper setter must put note (ii) in the question paper.

The syllabus has been divided into four units.

There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 10 short answer type questions carrying 20 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice of attempting one question from each Unit – 4 in all. Each question will carry 15 marks.

UNIT-I

Introduction : Nature, Scope and Methodology of Econometrics, Specification of an Econometric Model, Nature of data used in Estimation.

Single Equation Estimation : Method of Ordinary Least Squares (OLS) for Simple and Multiple Regression Models, Properties of the Least Squares Estimators, the Gauss-Markov Theorem, Tests of significance of OLS estimates and confidence intervals, Maximum Likelihood Estimation (MLE), Application and Uses of Dummy Variables (both as dependent and independent), Criteria for Model Selection.

UNIT-II

Problems with OLS Estimation : Multicollinearity, Specification Errors, Heteroscedasticity, Autocorrelation, Aitken's Generalized Least Squares (GLS) Method of Estimation, Estimation from Grouped Data, Errors of Measurement.

Dynamic Econometric Models : Autoregressive and Distributed Lag Models, Kyock's and Almon's Methods of Estimation, Rational Expectations Models, Granger and Sim's Causality Tests.

UNIT-III

Simultaneous Equation Models : Description, Identification Problem, Rank and Order Conditions.

Single Equation Methods of Estimation : Indirect Least Squares (ILS), Two Stage Least Squares (2 SLS), and Limited Information Maximum Likelihood (LIML); System Methods of Estimation : Three Stage Least Squares (3 SLS).

UNIT-IV

Time Series Analysis : Stationarity, Unit roots, Co-integration, spurious regression, Dickey-Fuller test, Engle-Granger test, Random walk models, Forecasting with ARIMA models, Box-Jenking Methodology; Vector auto-regression : Uses and Problems.

Panel Data Techniques : Random Coefficient; models; Fix effects models; Random effect model.

BOOKS RECOMMENDED :

Amemiya, T. (1985)	:	Advanced Econometrics, Harvard University Press, Cambridge, Mass.
Baltagi, B. H. (1998)	:	Econometrics, Springer, New York.
Box, G. and G. Jenkins (1984)	:	Time Series Analysis : Forecasting and Control, 2nd Ed. San Francisco : Holden Day.
Goldberger, A. S. (1998)	:	Introductory Econometrics, Harvard University Press, Cambridge Mass.
Gujarati, Damodar (2002)	:	Basic Econometrics (4th Edn.), McGraw Hill.
Intriligator, M.J. and Hasiao Bodkin (1996)	:	Econometric Models : Techniques and Applications, (2nd Edn.) Prentice Hall.
Johnston, J. and D. Nardo (1997)	:	Econometric Methods (4th Edn.), McGraw Hill.
Judge, G.C. Hill, W. Griffiths and T. Lee (1985)	:	Theory and Practice of Econometrics, New York : John Wiley and Sons.
Krishna, K. L. (1997)	:	Econometric Applications in India, Oxford University Press.

Koutsoyiannis, A. (1978)	:	Theory of Econometrics, Macmillan Publishers.
Maddala, G. S. (1997)	:	Econometrics, McGraw Hill, New York.
Mills, T. (1990)	:	Time Series Techniques for Economists, New York : Cambridge University Press.
Pesaran, M. H. & Peter Schmidt (Ed.) (1999)	:	Handbook of Applied Econometrics, Vols. I & II, Oxford University Press.
Pindyck, R. S. and Rubinfeld, D. L. (1998)	:	Econometric Models and Economic Forecasts (4th Edn.), McGraw Hill.
William, Green (2000)	:	Econometric Analysis (4th Edn.), Prentice Hall.

PAPERS III & IV OPTION (ii) : MATHEMATICAL ECONOMICS

100 Hours

This course is designed to equip students to understand the economic concepts and theories by using mathematical tools and techniques to refine the verbal logic. The use of calculus has permitted formulation of economic problems in multivariable mode and yield valuable insight about optimizing human behaviour. Modern algebraic tools allow convenient handling of simultaneous equations in the context of linear programming, game theory and input-output analysis.

Mathematical economics deals with various applications of mathematical tools and techniques in defining and developing economic relationships. This course has been accordingly designed to include various mathematical techniques/methods/models related to the different parts of economic theory like consumer theory, theory of production, pricing, trade cycles, growth models etc. The paper covers important aspects of microeconomics, macroeconomics and development theory and the various modules have been accordingly designed.

The time-tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars.

The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Instructions for the Paper-setters and candidates :

(i) The theory question paper will be of 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

The paper setter must put note (ii) in the question paper.

The syllabus has been divided into four units

There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 10 short answer type questions carrying 20 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice of attempting one question from each Unit – 4 in all. Each question will carry 15 marks.

UNIT-I

The Nature of Mathematical Economics and Ingredients of a Mathematical Model.

Theory of Consumer Behaviour : Utility Function-Behavioural postulates, nature and existence; Separable and additive utility functions; Homogeneous and homothetic utility functions; Demand function; Compensated demand function; Cardinal and Ordinal utility maximization; Slutsky equation for two and more commodities : Traditional and modern derivation and its interpretation; Concept of elasticities – generalization to n variable case; linear expenditure function and indirect utility function; Roy's identity; Expenditure function; Consumer's surplus; Choice under risk, Von Neuman utility function; Intertemporal utility maximization; Revealed preference theory—the weak and strong axioms, integrability problem; Composite commodity theorem.

Production Analysis : Homogeneous and Homothetic production functions; Elasticity of substitution; Cobb-Douglas, CES and VES production functions; Output and profit maximization, cost minimization; Duality in production; Input demand functions; Adding up Theorem; Analysis of joint profit maximisation and multi-product firm; Production possibility curve.

UNIT-II

Market Equilibrium : Price and Output determination under perfect competition, Monopoly and price discrimination, oligopoly and duopoly—the Cournot-Nash equilibrium, Collusion and Stackleberg model; Commodity market and factor market equilibrium; Existence and uniqueness of equilibrium; stability of equilibrium; Dynamic equilibrium with lagged adjustment; Cobweb model. Multi-market equilibrium, Walras's law and the existence of general equilibrium (without proof) and conditions for stability of equilibrium.

UNIT-III

Macro Economics : The Classical and Keynes' system; Hicks-Hansen version of the Keynesian model; fiscal policy and the balanced budget theorem; liquidity trap and the efficiency of monetary policy; Comparative statics in the Classical and Keynesian models.

Static and dynamic multiplier; Accelerator.

Trade Cycles : General concepts, Hicks-Samuelson multiplier-accelerator interaction model.

Growth Models : Harrod-Domar growth model and its stability; Solow model: Basic neo-classical growth model; Golden age problem; Technical progress and stability of neo-classical models. Optimal growth and golden rule of capital accumulation. Endogenous growth.

UNIT-IV

Input-Output Analysis : Static open and closed models, viability conditions; Activity analysis interpretation of input-output model; Aggregation problem; Substitution theorem; Cost and price determination; Dynamic Leontief system.

Linear Programming : Activity analysis interpretation of a linear program; Simplex and the revised simplex method; Dual and its economic interpretation; Complimentary slackness conditions; Duality theorems; Applications of linear programming in economics.

Game Theory : Two person zero sum non co-operant game, pure and mixed strategies; linear programming formulation of a matrix game; Methods of solving a game problem; Fair game and conversion of linear programming problem into a game problem. Introduction to non-constant sum games; Prisoner's dilemma.

BOOKS RECOMMENDED :

Abel, A., Bernanke, and Mcnabb (1998) : Macroeconomics, Addison Wesley, Massachusetts.

Allen, R. G. D. (1974) : Mathematical Economics, Macmillan, London.

Allen, R. G. D. (1974) : Mathematical Analysis for Economists, Macmillan Press & ELBS, London.

Allen, R. G. D. (1967) : Macro Economic Theory, Macmillan, London.

Baumol, W. J. (1978) (4th Edn.) : Economic Theory and Operations Analysis, Prentice Hall of India, N. Delhi.

Dorfman, R. Samuelson, P. and Sallow, R. (1968) : Linear Programming and Economic Analysis, McGraw Hill, Kogakusha Ltd., Tokyo.

Dornbusch, R. and Fisher, S. (1981) : Macroeconomics, McGraw Hill, N. Delhi.

Hamberg, D. (1971) : Models of Economic Growth, Harper & Row, New York.

Henderson, J. M. & Quandt, R. B. (1980) : Micro Economic Theory : A Mathematical Approach (3rd ed.), McGraw Hill, New Delhi.

Lancaster, V. (1965) : Mathematical Economics, Rand McNally College Pub.

Silberberg, E. (1978) : The Structure of Economics – A Mathematical Analysis, McGraw Hill, New York.

Varian, H. R. (1992) : Microeconomic Analysis, W. W. Norton, New York.

Nehar, Phillips, A. (1971) : Economic Growth and Development : A Mathematical Introduction, John Wiley, New York.

Madnani, G. M. K. : Mathematical Economics, IBH Publishing Co. Pvt. Ltd., New Delhi.

OTHER BOOKS :

Chiang, A. C. (1984) : Fundamental Methods of Mathematical Economics (3rd edn.), McGraw Hill, New York.

Chung, J. W. (1993) : Utility and Production : Theory and Application, Basil Blackwell, London, New York.

Green, H. A. J. (1971) : Consumer Theory, Penguin, Harmondsworth.

Jha, R. (1991) : Contemporary Macroeconomic Theory and Policy, Wiley Eastern Ltd., New Delhi.

Jones, H.G. (1976) : An Introduction to the Modern Theory of Economic Growth, McGraw Hill Kogakusha, Tokyo.

Mahesh Chand & Anand, V. K. (1981) : Economic Theory – A Mathematical Approach, Kitab Mahal, Allahabad.

Nash, J. F. (1996) : Essays on Game Theory, Cheltenham, U.K.

Quirk, J. and Saposnik, R. (1968) : Introduction to General Equilibrium Theory and Welfare Economics, McGraw Hill, New York.

Sen, A. K. (ed.) (1990) : Growth Economics, Penguin, Harmondsworth.

Solow, Robert M. (2000) : Growth Theory – An Exposition, Oxford University Press, Oxford.

Taha, H. S. (1997) : Operations Research – An Introduction (6th edition), Prentice Hall of India Pvt. Ltd., New Delhi.

Leontief, W. W. (1966) : Input-Output Economics, Oxford University Press, New York.

Todaro, M. P. (1971) : Development and Planning Models and Methods, Oxford University Press, Oxford.

PAPERS III & IV OPTION (iii) : INTERNATIONAL ECONOMICS

100 Hours

The course intends to provide a deep understanding about the broad principles and theories, which tend to govern the free flow of trade in goods, services and capital – both short-term and long-term – at the global level. The contents of the paper spread over different modules, lay stress on the theory and nature of the subject which, in turn, will greatly help them to examine the impact of the trade policies followed both at the national and international levels as also their welfare implications at macro level and the distribution of gains from trade with particular reference to India. Finally, it prepares the students in examining contents in Indian context.

The time-tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars.

The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Instructions for the Paper-setters and candidates :

(i) The theory question paper will be of 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.
The paper setter must put note (ii) in the question paper.

The syllabus has been divided into four units.

There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 10 short answer type questions carrying 20 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice of attempting one question from each Unit –4 in all. Each question will carry 15 marks.

UNIT-I

The Pure Theory of International Trade : Theories of absolute advantage, comparative advantage and opportunity costs. Heckscher Ohlin theory of trade, factor price equalization. Empirical testing of Heckscher Ohlin theory. Impact of dynamic factor in international trade (changes in tastes, technology and factor endowment), (The Rybczynski Theorem and immisering growth). Gains from trade and distribution. Terms of trade : Concepts and measurement of terms of trade. Secular deterioration hypothesis. New approaches to trade theory (The product cycle and economics of scale theory), preference similarity hypothesis. Causes of emergence and measurement of intra-industry trade.

UNIT-II

Theory of Tariffs : Effects of tariffs on balance of payments, terms of trade, national income, consumption, output and income distribution. Emergence of and the political economy of Non-tariff barriers. Regionalism and multilaterism. Trade restrictions and economic development of Southern countries. Optimum and effective rate of tariffs. Various forms of economic integration : The Theory of customs union. Static and dynamic effects of customs union. Problems and prospects of forming a customs union in developing areas with reference to Asian region.

UNIT-III

The Balance of Payments and Adjustments : Concepts and components of balance of payments. Causes of dis-equilibrium in the balance of payments. The process of adjustment in the Balance of Payments under Gold Standard, fixed exchange rate and flexible exchange rate systems. Elasticity and absorption approaches to balance of payments. Monetary and fiscal measures for adjustment in balance of payments dis-equilibrium, both under flexible and fixed exchange rates systems. Monetary approach to the balance of payments. Foreign trade multiplier with and without foreign repercussions.

UNIT-IV

Role of Multinational corporations in developing countries with special reference to India. The Euro-dollar and eurocurrency market and developing countries. International economic institutions. Functions and achievements of GATT/WTO. Impact of WTO on different sectors of the Indian Economy. UNCTAD broad outcome of VI, VII, VIII Rounds and their relevance for developing countries. Anatomy of the new international economic order. **TRIPS** and **TRIMS**, MFA and WTO. The Theory of optimum currency areas.

The Bretton Woods System : Its working and reasons for its collapse. Problems of international liquidity. Proposals for international monetary reforms and IMF. Developing Countries and the reform of International monetary system. East Asia : Success and crisis.

Recent trends towards trade liberalisation policies in India and their implications for trade and growth.

BOOKS RECOMMENDED :

Aggarwal, M. R. (2001) : International Institutions and Development in Developing Countries, Deep & Deep Publications, New Delhi.

Anne-o- Krueger (1984) : Trade Policies in Developing Countries, Published in R. W. Jones and P.B. Kenen (ed.), Handbook of International Economics, Vol. I, 1984.

Bhagwati, J. (1988) : Protectionism, Cambridge University Press, USA.

Brahmananda, P. R. (1982) : The IMF Loan and India's Economic Future, Himalaya Publishing House, Bombay.

Brahmananda, P. R. & V. R. Panchamukhi (ed.), 1987 : The Development Process of the Indian Economy, Himalaya Publishing House, New Delhi.

Chacholiades, M. (1980) : International Trade : Theory and Policy, McGraw Hill, Kugakusha, Japan.

Clement, Mpfister, R. L. & Rothwell, K. J. (1976) : Theoretical Issues in International Economics, Constable Publications, London.

Crockett, A. (1982) : International Money : Issues and Analysis, The ELBs and Nelson (UK).

Nayyar, Deepak (1976) : India's Exports and Export Policies in the 1960's, Cambridge University Press.

Salvatore, Dominick (1987) : The New Protectionist Threat to World Welfare.

Finance and Development (March, 2000) : Surmounting the Challenges of Globalization, IMF.

Helpman, H. & Krigman, P. R. (1985) : Market Structure and Foreign Trade, MIT Press.

Whally, John (1985) : Trade Liberalization among Major Trading Areas, Cambridge MIT Press.

Kenen, P. B. (1994) : The International Economy (3rd ed.).

Kindleberger, C. P. (1991) : International Economics (8th ed.), India Traveller Books Sellers, New Delhi.

Kindleberger, C.P. (1996) : Maxias, Panics and Crashes –A History of Financial Crisis.

Lall, S. & Streeten, P. (1977) : Foreign Investment, Traditional and Developing Countries.

Singh, Manmohan (1964) : India's Export Trends and the Prospects for Self Sustained Growth.

Martin, Feldstein (1990) : International Economic Co-operation.

Nagy, Bela Csikos, and Young, D. G. : East-West Economic Relations in the Changing Global Environment.

Ferguson, Nial (2001) : The Cash Nexus Money and Power in Modern World (1700-2000).

OECD (1985) : Costs and Benefits of Protection, Paris, OECD.

Panchmukhi, V. R., Raipuria, K. M. and Tandon R. (ed.)(1987) : Money and Finance in a World Economic Order, Indus Publishing Co., New Delhi.

Panchmukhi, V. R. (2000) : WTO and India, Challenges and Perspectives, RIS Discussion Papers.

Paul, R. Krugman & Maurice Obstfeld (2000) : International Economics Theory and Policy (5th ed.), Addison-Wesley Series in Economics.

Alibev, R.Z. (1987) : The International Money Game, Basic Books, New York.

Solomon, Robert (1982) : The International Monetary System (1945-1981), Marker and Row Publishers, New York.

Salvatore, D. (1996) : Theory and Problems of International Economics (7th ed.), Prentice Hall, New York.

Salvatore, D. (1997) : International Economics, Prentice Hall, New York.

Schwartz, A. J. (1987) : Money in Historical Perspective, Chicago University Press, Chicago.

Sodersten, B. O. (3rd ed.) (1999) : International Economics, The Macmillan Press Ltd., London.

The World Bank (2002) : Globalization, Growth and Poverty : Building an Inclusive World Economy, World Bank Policy Research Paper, Oxford University Press, New York .

PAPERS III & IV OPTION (iv) : ECONOMICS OF PUBLIC FINANCE

100 Hours

Role and functions of the Government in an economy have been changing with the passage of time. The term 'Public Finance' has traditionally been applied to the package of those policies and operations which involve the use of tax and expenditure measures while budgetary policy is an important part to understand the basic problems of use of resources, distribution of income, etc. There are vast array of fiscal institutions – tax systems, expenditure programmes, budgetary procedures, stabilization instruments, debt issues, levels of government etc., which raise a spectrum of issues arising from the operation of these institutions. Further, the existing of externalities, concern for adjustment in the distribution of income and wealth, etc. require political processes for their solution in a manner which combines individual freedom and justice. This paper combines a thorough understanding of fiscal institutions with a careful analysis of the issues which underline budgetary policies in general and Indian experience in particular.

The time-tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars.

The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Instructions for the Paper-setters and candidates :

(i) The theory question paper will be of 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.
The paper setter must put note (ii) in the question paper.

The syllabus has been divided into four units.

There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 10 short answer type questions carrying 20 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice of attempting one question from each Unit –4 in all. Each question will carry 15 marks.

UNIT –I

Issues Relating to the Economic Activities of the Government/Public Sector. Instruments of Public Finance : Concept and the role of Budgets (short-term demand management, redistribution of income and wealth).

Welfare Foundations of Public Economics : Pareto Optimality, Market Failure, Distinction between Private and Public Goods. The Optimal provision of Public Goods, Lindhal Equilibrium, Free Rider Problem. The Pure Theory of Public Goods by Samuelson.

UNIT-II

Theory of Taxation : Various approaches of equity in taxation. Benefit Principle, Lindahl and theory. Ability to Pay Approach.

Tax Incidence : Partial and General Equilibrium Incidence Analysis, excess burden, effects on work effort, savings, investment and growth.

UNIT-III

Public Expenditure : Growth and Structural changes, economic and political—Wagner's Law, Peacock and Wiseman's hypothesis, Criteria for Expenditure Evaluation : Valuation of Benefit and Cost, Choice of Discount rate.

Macro Analysis of Public Debt : Functional Finance and the Controversy relating to the burden of public debt and its shifting, Public Borrowing, Debt Management, Deficit Financing.

UNIT-IV

Fiscal Policy in a Macro Perspective in India : Fiscal Deficit and Inflation, Indian Tax System, Pattern of Govt. Expenditure, Public Debt Management, Problem of Tax Evasion and Parallel Economy.

Issues on Federal Finance in India : Devolution Criteria, Peculiarities and Problems.

BOOKS RECOMMENDED :

Brown, C. V. and Jackson, P. M. (1990) : Public Sector Economics, 4th edition, Basil Blackwell, Oxford.

M. Govinda Rao and Tapas Sen (1998) : Financial Federalism in India, Mcmillan, Delhi.

Hemalata Rao (1994) : Federal State Financial Relations : Theories and Principles, Ashish Publishing House.

Arun Kumar (1998) : Black Economy in India, Penguin Books, New Delhi.

Musgrave, R. A. & Musgrave, P. B. (Latest Edition) : Public Finance in Theory and Practice, McGraw Hill International eds.

Houghton, R. W. (ed.) (1970) : Public Finance, Harmondsworth, Penguin Books.

Atkinson, A. B. and Stiglitz, J. E. : Lectures on Public Finance, Maiden head, U.K.

Peacock, and Straw, G. K. (1970) : The Economic Theory of Fiscal Policy, George Allen and Unwin, London.

Chelliah, R. C. (1996) : Sustainable Growth, Essays on Financial and Fiscal Sector Reforms, Oxford University Press.

Raghabendra Jha (1999) : Modern Public Economics, Routledge.
Govt. of India, Ministry of Finance : 9th, 10th and 11th Finance Commission Reports.
Govt. of India, Ministry of Finance (1988) : Sarkaria Commission Report on Centre-State Financial Relations.

PAPERS III & IV OPTION (v) : ECONOMICS OF MONEY AND BANKING

100 Hours

Money and banking constitute important components towards understanding of economics. A clear understanding of the operations of money and banking and their interaction with the rest of the economy is essential to realize how monetary forces operate through a multitude of channels – market, non-market, institutions and among others, the state. The operation of financial markets and their regulations are to be studied to appreciate their key-role in an economy, especially after the far reaching banking and financial sector reforms in India and elsewhere. The present course is designed to acquaint the students fully with the changing role of financial institutions in the process of growth and development. Accordingly, the paper on 'Economics of Money and Banking' is an optimal integration of monetary theory, banking and non-banking financial institutions, which combines with itself a systematic discussion of the theory, institutions and policy with special reference to India.

The time-tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars.

The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Instructions for the Paper-setters and candidates :

(i) The theory question paper will be of 80 marks and 20 marks will be for internal assessment.
(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.
The paper setter must put note (ii) in the question paper.

The syllabus has been divided into four units.

There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 10 short answer type questions carrying 20 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice of attempting one question from each Unit – 4 in all. Each question will carry 15 marks.

UNIT-I

Money : Post war controversy on the definition of Money and Money supply. Money and Near Money Assets. Theoretical and empirical attempts to define money. Ordinary money and high powered money. Inside money and outside money.

Demand for Money : Classical theory, Keynes' Liquidity Preference Approach, Patinkin and the Real Balance Effect, Friedman's theory, Empirical Evidence on Demand for Money.

Supply of Money : Components and Sources of Money Supply, A simple theory of money supply. Money multiplier process, Determinants of money supply. Analysis of money supply in India - Reports of different RBI working groups.

UNIT-II

Monetary Policy : Goals, targets, indicators, instruments and effectiveness of monetary policy. Role, objectives and limitations of the monetary policy in developed and developing countries. Transmission mechanism of monetary policy, Rules versus Discretion, Monetarism, Debate on Neutrality of Money Monetary targeting—its relevance in India.

Term Structure of Interest Rates : Theories of term structure of interest rates, Structure of interest rates in India. Appropriate interest rate policy.

UNIT-III

The Central Bank : Role and functions, Techniques of Monetary Control, the Changing Context of Monetary Policy in India, Autonomy of Central Banks—RBI Autonomy and Monetary Management.

Commercial Banks : Role and functions, major developments in Commercial Banking in India since independence. Cooperative Banks in India – major developments in recent years.

Non-bank Financial Intermediaries : Distinction between Banks and NBFIs. Their growth, structure and impact on India's economic development. Control and Regulation of NBFCs by RBI.

UNIT-IV

Financial Markets : Money Market and Capital Market – Main Functions and Structure. Efficiency of Money Market. Structure of Indian Money and Capital Market**.** SEBI : Its impact on the working of Capital Market. Financial Sector Reforms in India and Agenda for Future. Issues regarding Convertibility–Implications of Tarapore Committee Report.

International Monetary System : International Liquidity and the IMF. India and the IMF. European Monetary System—Main features of Maastricht Treaty—Post Maastricht developments. Reforms in international monetary system for developing countries.

BOOKS/JOURNALS RECOMMENDED :

Aschheim, J. and Hsieh, C. (1969)	:	Macro Economics : Income and Monetary Theory. Columbus, Ohio, Charley E. Merrill.
Benjamin, M. Friedman and Frank H. Hahn (1990)	:	Handbook of Monetary Economics (North Holland).
Benett, T. McCallum (1996)	:	International Monetary Economics, Oxford University Press, Oxford.

Bhole, L. M. (1999) : Financial Institutions and Markets, Tata McGraw Hill Ltd., New Delhi.

Bhole, L. M. (2006) : Indian Financial System, Chugh Publications, Allahabad.

Carr J. and M. Darby (1981) : The Role of Money Supply Shocks in Short Run Demand for Money, Journal of Monetary Economics, Vol. 8.

Fry, M. J. (1982) : Money, Interest and Banking in Economic Development, John Hopkin's University Press.

Gupta, S. B. (1995) : Monetary Planning for India, Oxford University Press, Oxford.

Gupta, S. B. (1982) : Monetary Economics—Institutions, Theory and Policy, S. Chand and Company, N. Delhi.

Gurley, J. G. and Shaw, E. S. (1960) : Money in a Theory of Finance, Brooking Institution, Washington.

Heller, H. Robert (1968) : International Monetary Economics, Prentice Hall, India.

James Tobin (1965) : Money and Economic Growth, Econometrica, Volume 33.

Jadhav, N. (1964) : Monetary Economics for India, Macmillan India Ltd., New Delhi.

Jadhav, N. (1996) : Challenges to Indian Banking (Ed.), Macmillan India Limited, New Delhi.

Khan, M. Y. (1996) : Indian Financial System, Tata McGraw Hill, New Delhi.

Kulkarni, G. (1999) : Modern Monetary Theory (Macmillan), Delhi.

Laidler, David (1972) : Demand for Money : Theories and Evidence, Allied Publishers, New Delhi.

Laidler, David (1999) : The Foundations of Monetary Economics, Volume II, Edward Elgar Publishing Ltd.

Mckinen, G. E. (1978) : Money, The Price Level and Interest Rates, Prentice Hall of India, New Delhi.

Mishkin, S. F. (1998) (5th ed.)	:	The Economics of Money, Banking and Financial Markets, Harper Colins Publishers, New York.
Newlyn, W.T. (1978)	:	Theory of Money (3rd Edition), Clarendon Press Oxford, London.
Patinkin, Don	:	Money, Interest and Prices – An Integration of Monetary and Value Theory, New York, Harper and Row, 2nd Edition.
Prasad, K. N. (2001)	:	Developments in India's Financial System, Sarup and Sons, New Delhi.
Peter, J. Montrel (1991)	:	Transmission Mechanism of Monetary Policy for LDCs, IMF Staff Papers, Vol. 38.
R. Dornbusch and A. Giovannini	:	Monetary Policy in the Open Economy, Chapter 23, Handbook of Monetary Economics.
Rangarajan, C. (1999)	:	Indian Economy : Essays on Money and Finance, Universal Book Stall, New Delhi.
Reddy, Y. V. (2000)	:	A Review of Monetary and Financial Sector Reforms in India – A Central Banker's Perspective, Universal Book Stall, New Delhi.
Reserve Bank of India	:	Report of the Committee to Review the Working of the Monetary System in India; Chakravarty Committee Report, Mumbai.
	:	Report of the Committee on the Financial System.
	:	Report of the Committee on Capital Account Convertibility.
	:	Annual Report (Recent Issues).
	:	Report on Currency and Finance (Recent Issues).
	:	RBI Occasional Papers.
		Report of the Working Group on Money Supply : Analysis and Methodology of Compilation, RBI Bulletin, July, 1998.
Schwartz, A. J. (1987)	:	Money in Historical Perspective, Chicago University Press, Chicago.
Trevor M. Sikorki (1996)	:	Financial Liberalization in Developing Countries, Edward Elgar.

OR

PAPERS III & IV OPTION (v) : ECONOMICS OF FINANCIAL MARKETS AND INSTITUTIONS

100 Hours

The positive and significant role of financial institutions in the process of growth and development has been very well recognized in the literature and indeed has become more important during the last two decades as the financial systems of different countries have become integrated in the process of globalization. India is no exception and has taken far reaching measures since 1991 in this direction. It is, therefore, essential that the students of economics should be well conversant with the theory and practice of different financial institutions and markets to understand and analyse the interconnection between the monetary forces and real forces, their developmental role and limitations in shaping and influencing the monetary and related policies both at the national and international levels.

The time-tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars.

The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Instructions for the Paper-setters and candidates :

(i) The theory question paper will be of 80 marks and 20 marks will be for internal assessment.
(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.
The paper setter must put note (ii) in the question paper.

The syllabus has been divided into four units.

There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 10 short answer type questions carrying 20 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice of attempting one question from each Unit – 4 in all. Each question will carry 15 marks.

UNIT-I

Financial Markets and Financial Instruments – The place of Financial Markets in the Economy – A framework for macro economic analysis of financial and non-financial flows – The relation between stocks and flows – Rate of Interest – Exchange rates. Structure and Functions of Financial Institutions.

Introduction to Corporate Finance – Forms of Business Organization – Goals of Financial Management and Basic Concept of Accounting, The Cost of Capital and the Capital Structure.

UNIT-II

The Supply of Securities, Regulations governing Supply of Securities – General Characteristics of Securities – Government bonds – Index Linked Bonds – Corporate Securities, Mutual Funds, Money Market Funds.

The Demand for Securities – The time dimension – present value and duration, yields on zero-coupon bonds – The term structure of interest rates (with special reference to India). Measurement of risk, expected utility approach, Mean variance portfolio theory, portfolio diversification.

UNIT-III

Efficiency of Securities Markets, The Efficient Market Hypothesis (EMH), mean, semi-strong and strong form of EMH, The Capital Asset Pricing Model (CAPM), the simplest form, estimating betas, validity of CAPM, the Arbitrage Pricing Theory (APT), Introduction to Stock Indices.

Securities Analysis and Market Efficiency – Macro economic developments and securities markets. Performance of Securities Markets and Securities Markets – Industry growth, structure and Firm specific factors – Uses and pitfalls of Price/Earnings Ratios.

UNIT-IV

Introduction to Derivatives, Futures and Options, commodity and finance futures, Hedgers, Speculators and Market Equilibrium, Option Pricing Theory, Binomial approach, Black-Sholes Option pricing model. The put-call parity theorem, other option models, Currency, Equity and Commodity Swaps.

Regulation of Financial Markets – The ethics of finance and the economic function of financial markets, Securities and Exchange Board of India (SEBI) – Role and Functions, Recent Reforms in the Indian Financial Market.

BOOKS RECOMMENDED :

Bhole, L. M. (2006)	:	Financial Institutions and Markets, Tata McGraw Hill, New Delhi.
Brealey, R. and Myers, S. (1997)	:	Principles of Corporate Finance, New York, McGraw Hill.
Govt. of India	:	Narasimhan Committee Report, 1992.
Houthakker, H. S. and Williamson, P. J. (1996)	:	Economics of Financial Markets, Oxford University Press.
Luenberger, David (1998)	:	Investment Science, Oxford University Press.
Fabozzi, Frank J, Modigliani Franco, and Ferri Michael (1998)	:	Foundation of Financial Markets and Institutions, Pearson Ed.
Khan, M. Y. (2000)	:	Indian Financial System, Tata McGraw Hill, New Delhi.
Reserve Bank of India	:	Reports on Currency and Finance Annual.

PAPERS III & IV OPTION (vi) : POLITICAL ECONOMY

100 Hours

The objective of this paper is to expose the students to alternative perspectives on the study of change, analysis of Economic behaviour, distribution and exercise of power.

The time-tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars.

The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Instructions for the Paper-setters and candidates :

(i) The theory question paper will be of 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

The paper setter must put note (ii) in the question paper.

The syllabus has been divided into four units.

There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 10 short answer type questions carrying 20 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice of attempting one question from each Unit –4 in all. Each question will carry 15 marks.

UNIT-I

Meaning, Nature and Evolution of Political Economy. Essential features of Neo-classical economics and its critique. Some Concepts and Theories : The concept of power, Theories of class and class formation. Theories of State, Theory of economic policy. The concept of dialectical materialism.

UNIT-II

Historical Social formations and emergence of capitalism. The process of primitive accumulation under capitalism. Nature of Capitalist Development : Schumpeters formulation.

UNIT-III

Monopoly, Capitalism and Imperialism. Neocolonialism : The highest stage of Imperialism. Political Economy of Development and Underdevelopment : Theories of Paul Baran, G. Frank and A. Emmanuel.

UNIT-IV

Marx and Classical Economists. Marx's Method of Analysis. Transformation problem. Falling Rate of Profit. Theory of Reserve Army of Labour. Marx's Theory of Alienation. Marx's Theory of Money. Asiatic Mode of Production. Marx on Development and Under-development. Contemporary Reference of Marx.

BOOKS RECOMMENDED :

John Eaton (1973)	:	Political Economy, International Publishers, New York.
R. Blackbun (Ed.), 1973	:	Ideology in Social Sciences.
--do--	:	Reading in Critical Social Theory Collins,73, London.
M. Blaug (1978) (Ed.)	:	Economic Theory in Retrospect, Vikas Publishing House, New Delhi.
Tom Bottomore, 1985	:	A Dictionary of Marxist Thought, Basil Blackwell Publishers, England.
Tom, Bottomore ed., 1988	:	Interpretation of Marx, Oxford, Basil Blackwell,1988.
Paul M. Sweezy (1946)	:	Theory of Capitalist Development, Dennis Dolson Ltd., London.
M. C. Howards & J. E. King (Ed.) (1976)	:	The Economics of Marx, Penguin Books Ltd., Harmondsworth, England.
Joan Robinson (1966) 2nd edn.	:	An Essay on Marxian Economics, Macmillan, London.
Karl Marx,1976 VI, 1978 VII	:	Capital, Vols. I, II, III, Penguin Books, Harmondsworth, England
Karl Marx (1973	:	Grundrisse.
G. C. Wang (1977)	:	Fundamentals of Political Economy.
B. N. Ghosh (1990)	:	Political Economy : A Marxist Approach (Macmillan), Delhi.
Maurice Cornforth, 1955-56	:	Dialectical Materialism : An Introduction, London, Lawrence 1955-56, 3 Vol.
B. Hindess and P. S. Hirst (1975)	:	Pre-capitalist Modes of Product.
Tom Kemp (1967)	:	Theories of Imperialism, Lond., Dolison.
Owen R. and Sutecliffe, Bole (1975)	:	Studies in the Theory of Imperialism, London, Longman.
K. Nkrumah	:	Neocolonialism.
Jack Woodis	:	An Introduction to Neocolonialism.
Water Calvert (1982)	:	The Concept of Class.
B. N. Ghosh, et.al. (1991)	:	Five Great Economists (5 vols.), Deep & Deep Pub., Delhi.

C. A. Kregel (1973)	:	Reconstruction of Political Economy.
D. Mermelstein (1970)	:	Economics : Mainstream Readings and Radical Critique.
C. Myrdal (1932)	:	Political Element in the Development of Economic Theory.
R. Miliband (1969)	:	The State in Capitalist Society.
N. Poulantzas	:	Political Power and Social Classes.
N. Poulantzas (1978)	:	Classes in Contemporary Capitalism.
C. Wilber (Ed.) (1973).	:	Political Economy of Development and Underdevelopment.
H. Bernstein (Ed.) (1973) .	:	Underdevelopment and Development, Penguin Books Ltd., Harmondworth, England .
P.A. Baran (1957) .	:	Political Economy of Growth, Modern Reader Paperbacks, New York.
N. T. Uphoff and W. Illchman (Ed.) (1972)	:	Political Economy of Development, University of California Press.
Theodore Morgan (1975)	:	Economic Development .
D. L. Johnson & R. H. Chilcote	:	Theories of Development : Mode of Production or Dependency.
T. Wilson (1982)	:	The Political Economy of the Welfare State .
E. O. Wright (1979)	:	Class, Crisis and the State .
P. Coontz	:	Population Theories : An Economic Analysis.
Bhikhu Parekh (1982)	:	Marx's Theory of Ideology, Delhi.
P. K. Bardhan's Paper I John Roemer (1986)	:	Analytical Marxism.
Paul, M. Sweezy, 1979	:	A Crisis in Marxism, Monthly Review, June.

PAPERS III & IV OPTION (vii) : COMPUTER APPLICATIONS IN ECONOMIC AND STATISTICAL ANALYSIS

100 Hours

The direction towards the knowledge society shall be resting on a healthy economics and proper use of Information Technology. Economics being an empirical science, computers have emerged as the pivotal instruments of economic analysis, research and forecasting. Therefore, the future economists must be equipped with skills and tools based on computers, which this course shall provide to them. This will not only enhance their employability but also prepare them for the challenges of the future.

The time-tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars.

The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Instructions for the Paper-setters and candidates :

(i) The theory question paper will be of 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.
The paper setter must put note (ii) in the question paper.

The syllabus has been divided into four units.

There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 10 short answer type questions carrying 20 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice of attempting one question from each Unit –4 in all. Each question will carry 15 marks.

UNIT-I

Computer Fundamentals : Basic Introduction to Computer and its Components, Input/Output devices, concepts of primary/secondary storage, types of software, configuration of a personal computer.

Using DOS and Windows : Introduction to operating system, Functions and types of operating system, Fundamentals of DOS, Internal and external commands, batch files, ASCII and multiple files, Use of Editors like ED/EDIT; Windows fundamentals and administration, anatomy of a window, start menu, my computer, creating and using groups, customizing windows, installing a printer, using clipboard; Introduction to UNIX operating system.

UNIT-II

Fundamentals of MS-Word : Menus, Toolbars, Ruler, Scroll Bars, Status Bar; Creating, Saving, Importing, Exporting and Inserting Files; Formatting, Indents/Outdents, Lists, Tabs, Styles; Working with Frames, Columns, Pictures, Charts/Graphs, Forms, Tools, Equations and Macros.

Overview of Programming Languages : Compiler, assembler, interpreter; First generation languages to current state of language development, basic features of FORTRAN, writing simple programs in FORTRAN for correlation, regression, matrix inversion etc.

UNIT-III

Organisation and Manipulation of Data and Data Files (Introduction to MS Excel, Lotus 1-2-3) : Importing and Exporting of data files with special emphasis on creation and transformation of Cross-section and time series data; pooling of cross section and time series; panel data, concept of lags and leads, Use of macros and simple statistical, financial and mathematical functions; Use of simple graphics, Basics of using panel data, Reading in panel data, Unbalanced panels.

Statistical Package for Social Sciences (SPSS) : Creating data file, Correcting data, Assigning names and labels to variables, missing values, different types of variables, Saving and retrieving the data file, Using the menus to analyse the data, summarize, compare means, ANOVA models, correlation, regression, multiple regression, parametric (Pearson) and rank correlation, partial correlation, Chi-Square for 2 or more independent sample, selecting permanent/temporary subscripts, graphs in SPSS, merging data files in SPSS, Transforming files between SPSS and other programs; transformation of data and output (text) files to/from a word processor, transfer data files to/from other packages, transferring chart files.

UNIT-IV

Database Management : Data and Information, Definition of Database, Objectives of Database, Database Management System, Levels of Database, Data Storage and Retrieval, Architecture for a Database System, Data Languages, the Hierarchical Structure, The Network Structure, Relational Structure etc.

Introduction to System Models : An overview, Inventory Model, Progress Model, Writing-line Model, Markov Process Model, Forecasting Model, Linear Programming Model etc.

Internet : Use of E-mail and Internet; Downloading information related to economics, processing and presenting them through a project report.

BOOKS RECOMMENDED :

Ashok Arora and Akshay Bhatia	:	Information System for Managers, Excel Books, New Delhi.
Rajaraman, V. (1996)	:	Fundamental of Computer, Prentice Hall of India, New Delhi.
Majumdar, A. K. and P. Bhattacharjee (1996)	:	Database Management Systems, Tata McGraw Hills.
William E. Mayo and Martin Curiakala (1995) SPSS 7.5 Users Manual	:	Theory and Problems of Programming with Fortran. 77, McGraw Hill.
B. W. Kerningham	:	The Unix Programming Environment.
Cady, G. H. and Pat McGregor	:	Mastering the Internet, New Delhi.

PAPERS III & IV OPTION (viii) : ECONOMICS OF AGRICULTURE
OR
ECONOMICS OF INDUSTRY—

whichever the student has not offered in M.A.-I. The syllabi in this option will be the same as prescribed in M.A.-I

PAPERS III & IV OPTION (ix) : ECONOMICS OF ENVIRONMENT

100 Hours

This course is meant to provide some insights into the application of economic theory in the design and implementation of public policy related to the management of environment. The course finds roots in welfare economics, national income accounting, macro economic policies and trade and development.

The time-tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars.

The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Instructions for the Paper-setters and candidates :

(i) The theory question paper will be of 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

The paper setter must put note (ii) in the question paper.

The syllabus has been divided into four units.

There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 10 short answer type questions carrying 20 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice of attempting one question from each Unit – 4 in all. Each question will carry 15 marks.

UNIT-I

Scope of Environmental Economics : Basic Concepts, Environment sustainability, Development and Environment – Linkage and trade off, major environment problems, Introduction to Laws of thermodynamics, Renewable and non-renewable resources.

Environmental Externality : Market failure, Pigouvian, Coasian and Arrowian solutions; Non-convexities in production set, Uncertainty and choice of policy instruments (Graphical approach).

UNIT-II

Economics of Exhaustible Resources : Hotelling, Rule, Solow-Hartwick rule, Depletion Cost, Optimal pricing.

Economics of Renewable Resources : Model of biological population growth, fisheries, common property resources, forest conservation, community participation.

Economics of Sustainable Development : Sustainability rules, indicators of sustainability.

UNIT-III

Valuation Methods and Approaches : Evaluation criteria, Household production function method, Contingent Valuation Method (CVM), Hedonic price method, Travel cost and Random utility methods.

Environmental Instruments : Property regulations, Command and Control regulations; Economic, Fiscal and Financial Instruments (Pollution charges and Tradable permits).

UNIT-IV

Sources of Pollution and Environmental Policy in India : Sources and types of pollution, current states of pollution in India, Evolution of Environmental Policy in India : Air act, Water act, Pollution Control Policy, 1992.

Environmental Issues in the Global Context : Green House Gases (GHG) and Ozone depletion, Montreal protocol, Trade and Environment, GATT & Environment.

BOOKS RECOMMENDED :

Nick Hanley, Jason F. Shogen and Ben White (1999)	:	Environmental Economics in Theory and Practice (Macmillan India Ltd., New Delhi).
Charles D. Kolstad (2000)	:	Environmental Economics (Oxford Univ. Press, New York).
William Baumol and Wallane E. Oats (1998)	:	The Theory of Environmental Policy (2nd Edn.), Cambridge Univ. Press, Cambridge.
Daniel W. Bromly (Eds.) (1996)	:	Handbook of Environmental Economics (Blackwell Publishers Ltd.)
Hussen, A. H. (2000)	:	Principles of Environmental Economics, Routledge.
Mehta, S., Mundle, S. and Sarkar, U. (1997)	:	Incentives and Regulations for Pollution Control, (Sage Publishers).

PAPERS III & IV OPTION (x) : DISSERTATION

1. Dissertation may be submitted as one of the nine options, provided pupils who elect to do so satisfy the following conditions :

   (i) they are either on the rolls of the University Department or the affiliated Colleges or private candidates – having been college teachers with at least five years' teaching experience as on the last date for submitting (M.A. Part II) examination forms;

   (ii) they are allowed to write a dissertation as one of the options on a subject and under a supervisor, duly approved by the Board of Studies in Economics.

2. Pupils fulfilling the above conditions shall communicate to the University not later than the fifteenth of October preceding their Part II examination the precise title of their dissertation, the chapter-scheme of writing; a bibliography of books and articles.

3. In length, no dissertation ought to fall below 75 or exceed 125 pages. It should be on one and half spaced foolscap sheets neatly typewritten on both sides. The typescript must be properly bound in cloth or leather with the title of the Dissertation; name of the Candidate and Year of submission embossed on the cover and the spine.

4. The subject of thesis approved by the Board of Studies shall remain valid for four years only.

5. Stop using a blank sheet before the page carrying figure or table.

6. Try to insert figure/table in the text page itself (instead of using a fresh page for it each time).

7. Every candidate shall submit his thesis so as to reach the University at least 20 days before the commencement of Part II Examination. Delay in receipt of the thesis may be condoned upto April 30 of the year of examination but no thesis can be accepted after that date.

8. Submission and Evaluation of dissertation would be as per decision of the Academic Council Item No. XXII dated 28-04-2003 reproduced below :

   (i) Only such students will be eligible to take up dissertation as an optional paper in M.A. Part II who have secured at least 55% in aggregate in M.A. Part I Economics Examination.
   (ii) The weightage of written dissertation and viva-voce examination be 70 and 30 respectively.
   (iii) It will be essential for a candidate to secure at least 40% marks in viva-voce examination, i.e. 12 marks out of 30 to qualify in the dissertation.
   (iv) The viva-voce examination will be conducted at the Department of Economics, Panjab University, Chandigarh irrespective of the Centre/College from where the candidate has studied and submitted the dissertation.
   (v) These clauses pertaining to the evaluation of M.A. dissertation shall be applicable to such students who seek admission in M.A. Part I (Economics) i.e. from the academic session 2003-04 onwards.

9. Written Dissertation will be evaluated only by the External Examiner.

Published by : Professor A.K. Bhandari, Registrar, Panjab University, Chandigarh.